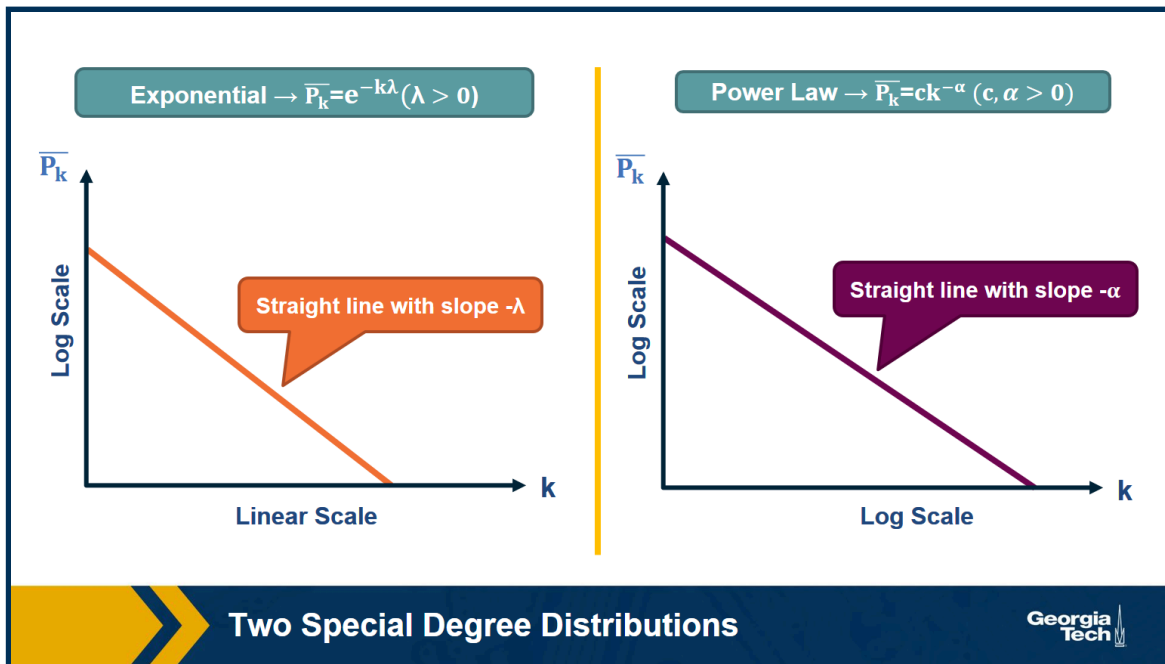


L3: Two Special Degree Distributions



The C-CDF plots are often shown using a logarithmic scale at the x-axis and/or y-axis. Here is why.

Suppose that the C-CDF decays exponentially fast. In a log-linear plot (as shown at the left in the image above), this distribution will appear as a straight line with slope $-\lambda$. The average degree in such networks is $1/\lambda$. The probability that a node has degree at least k drops exponentially fast with k .

On the other hand, in many networks, the C-CDF decays with a power-law of k . For example, if $\alpha = 2$, the probability to see a node with a degree at least k drops proportionally to $1/k^2$. In a log-log plot (as shown at the right), this distribution will appear as a straight line with slope $-\alpha$. As we will see later in this course, such networks are referred to as “**scale-free**” and they are likely to have nodes with much larger degree than the average degree.

Food for Thought

Show why the previous two distributions give straight lines when we plot them in a log-linear and log-log scale, respectively. Also show that, with the exponential degree distribution, the probability to see nodes with degree more than 10 times the average is about 1/10000 of the probability to see nodes with higher degree than the average. On the other hand, with the power-law distribution, the probability to see nodes with degree more than 10 times the average is 1/100 of the probability to see nodes with higher degree than the average (when $\alpha = 2$).